

# ASHA Worker Mobile App for Arogya-SMC: Complete Design Document

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## 1. Understanding Existing ASHA/Anganwadi Worker Apps

### Research Insights from Poshan Tracker & Similar Apps

Based on analysis of Poshan Tracker :

#### What They Do Well:

- **Real-time data entry** for growth monitoring, service delivery, beneficiary tracking
- **Offline mode** for field work with connectivity issues
- **Beneficiary profiles** for pregnant women, lactating mothers, children 0-6 years
- **Aadhaar-linked verification** to prevent duplicates
- **Growth charts** with WHO standards, color-coded risk levels (green/yellow/red)
- **Geo-tagging** of centers and visits
- **Photo-based verification** for attendance

#### What Users Hate (Critical Lessons):

The Poshan Tracker has a **2.8-3.0 star rating** with 127,000+ reviews . User complaints include:

| Problem                 | Real User Quote                                                                                                                                                        | Our Solution                                                     |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| <b>Frequent crashes</b> | "In the pursuit to smoothen and digitise the work of an AWW, this app has been total disaster. Frequent crash"                                                         | Robust error handling, crash reporting, thorough testing         |
| <b>GPS failures</b>     | "high failure to obtain correct GPS location" and "Despite being present at the centre, the location itself is shown as wrong"                                         | Graceful fallback, manual location entry option                  |
| <b>Slow performance</b> | "slow processing, issues are never ending"                                                                                                                             | Lightweight architecture, optimized queries                      |
| <b>Poor UI/UX</b>       | "Poor UI/UX design makes the Poshan Tracker app extremely frustrating to use. It's built for non-tech-savvy women, yet the interface is overly complex and confusing." | <b>Simple, intuitive design with large buttons, clear labels</b> |
| <b>Heavy app size</b>   | "The app is heavy, very slow, and consumes too much data"                                                                                                              | Target <40 MB, optimize assets                                   |

| Problem                            | Real User Quote                                                                                                 | Our Solution                                  |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <b>Old device issues</b>           | "The biggest problem... mobile phones which the government has given to the mobile users with very old system." | Support Android 5.0+, test on low-end devices |
| <b>Updates making things worse</b> | "with every upgrade, the number of problems only reach newer heights"                                           | Thorough regression testing                   |

## Key Features of Existing Systems

| Feature Category              | Details                                                                                               |
|-------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Beneficiary Management</b> | Pregnant women, lactating mothers, children 0-6 years, adolescent girls                               |
| <b>Services Tracked</b>       | Iron/folic acid intake, antenatal checkups, tetanus injections, nutrition supplements, postnatal care |
| <b>Growth Monitoring</b>      | Weight, height, WHO-standard growth curves, color-coded risk                                          |
| <b>Daily Operations</b>       | Attendance tracking, take-home rations (THR), hot cooked meals (HCM)                                  |
| <b>Verification</b>           | Aadhaar-linked profiles, photo-based verification                                                     |
| <b>Location</b>               | Center geo-tagging                                                                                    |

## 2. Technology Stack for ASHA App (100% Free)

### Core Framework

| Component        | Technology    | Justification                                                        |
|------------------|---------------|----------------------------------------------------------------------|
| <b>Framework</b> | Flutter 3.27+ | Single codebase, excellent offline support, works on low-end devices |
| <b>Language</b>  | Dart 3.5+     | Modern, type-safe                                                    |
| <b>IDE</b>       | VS Code       | Free, excellent Flutter plugins                                      |

### State Management & Architecture

| Component               | Technology                | Purpose                         |
|-------------------------|---------------------------|---------------------------------|
| <b>State Management</b> | Riverpod 2.5+             | Testable, no context dependency |
| <b>Navigation</b>       | GoRouter 14.0+            | Declarative routing             |
| <b>Architecture</b>     | MVVM (Clean Architecture) | Clear separation of concerns    |

### Local Storage & Offline

| Component             | Technology                   | Purpose                                                             |
|-----------------------|------------------------------|---------------------------------------------------------------------|
| <b>Local Database</b> | <b>Drift</b> (formerly Moor) | Type-safe SQLite, better than raw SQLite, excellent offline support |
| <b>Secure Storage</b> | flutter_secure_storage       | For tokens, credentials                                             |
| <b>Connectivity</b>   | connectivity_plus            | Detect network status                                               |
| <b>Work Manager</b>   | flutter_workmanager          | Background sync even when app closed                                |

## Networking

| Component                   | Technology          | Purpose                             |
|-----------------------------|---------------------|-------------------------------------|
| <b>HTTP Client</b>          | Dio 5.4+            | Interceptors, retry logic, timeouts |
| <b>API Client Generator</b> | Retrofit (for Dart) | Type-safe API calls                 |

## Utilities

| Component                 | Technology         | Purpose                        |
|---------------------------|--------------------|--------------------------------|
| <b>Location</b>           | geolocator 12.0+   | GPS for center geo-tagging     |
| <b>Camera</b>             | camera             | Photo capture for verification |
| <b>Image Picker</b>       | image_picker       | Select images                  |
| <b>Permissions</b>        | permission_handler | Runtime permissions            |
| <b>Push Notifications</b> | firebase_messaging | Receive alerts from MHO        |
| <b>Localization</b>       | easy_localization  | Marathi, Hindi, English        |
| <b>QR Scanner</b>         | mobile_scanner     | Optional Aadhaar scan          |
| <b>File Download</b>      | flutter_downloader | Download reports               |

## 3. Features & Functions for ASHA App

### Core Features (Must-Have for March 23)

| Feature                           | Description                                             | Priority |
|-----------------------------------|---------------------------------------------------------|----------|
| <b>F1: Secure Login</b>           | OTP-based login (simulate with demo mode for prototype) | HIGH     |
| <b>F2: Dashboard</b>              | Today's pending reports, recent alerts, quick actions   | HIGH     |
| <b>F3: Daily Syndromic Report</b> | Fever, cough, diarrhea counts with geotagging           | HIGH     |

| Feature                           | Description                                   | Priority |
|-----------------------------------|-----------------------------------------------|----------|
| <b>F4: Maternal Risk Flagging</b> | Pregnancy complications, high-risk indicators | HIGH     |
| <b>F5: Child Risk Flagging</b>    | Malnutrition flags, growth monitoring         | HIGH     |
| <b>F6: Environmental Flags</b>    | Stagnant water, poor sanitation               | HIGH     |
| <b>F7: Offline Storage</b>        | Save reports without internet, sync later     | HIGH     |
| <b>F8: Alert List</b>             | View outbreak alerts from MHO                 | MEDIUM   |
| <b>F9: Push Notifications</b>     | Receive real-time alerts                      | MEDIUM   |
| <b>F10: Sync Status</b>           | Visual indicator of pending/synced reports    | MEDIUM   |

### Secondary Features (Post-Prototype)

| Feature                  | Description                    |
|--------------------------|--------------------------------|
| <b>Beneficiary List</b>  | View families in assigned ward |
| <b>Growth Charts</b>     | Track child growth over time   |
| <b>Visit History</b>     | Past reports and visits        |
| <b>QR Scan</b>           | Quick beneficiary lookup       |
| <b>Report Generation</b> | Download monthly summaries     |

## 4. Data Model for ASHA Reports

Based on the workflow analysis from existing systems , here's what ASHA workers need to record:

```
class ASHAReporT {
    final String reportId;
    final String ashaWorkerId;
    final String wardCode;
    final DateTime reportDate;
    final DateTime submissionTime; // For sync tracking
    final bool isSynced;

    // Syndromic Surveillance
    final SyndromicData syndromic;

    // Maternal Health
    final MaternalHealthData maternal;

    // Child Health
    final ChildHealthData child;

    // Environmental
    final EnvironmentalData environmental;
}
```

```

    // Location
    final GeoLocation location; // Lat/lng of report
}

class SyndromicData {
    final int feverCount;
    final int coughCount;
    final int diarrheaCount;
    final int jaundiceCount;
    final Map<String, int> otherSymptoms; // Flexible for new outbreaks
}

class MaternalHealthData {
    final int pregnantWomenCount;
    final int highRiskPregnancyCount; // Flags
    final int ancVisitsConducted; // Antenatal care
    final int ironSupplementDistributed;
}

class ChildHealthData {
    final int childrenUnder5;
    final int malnourishedCount; // Severe acute malnutrition
    final int immunizationDue;
    final Map<String, double> growthMeasurements; // Weight, height
}

class EnvironmentalData {
    final bool stagnantWaterPresent;
    final bool poorSanitation;
    final bool garbageDumping;
    final String? otherRisks;
}

class GeoLocation {
    final double latitude;
    final double longitude;
    final double accuracy;
    final String? placemark; // Reverse geocoded ward
}

```

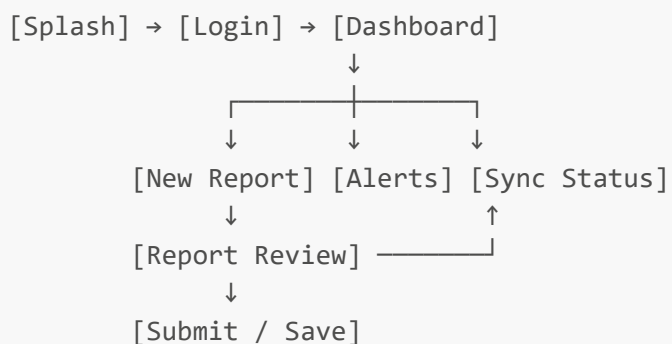
## 5. Pages / Screens Design

### Screen Overview

| Screen        | Route  | Purpose                                   |
|---------------|--------|-------------------------------------------|
| <b>Splash</b> | /      | Loading, check auth                       |
| <b>Login</b>  | /login | OTP-based login (demo mode for prototype) |

| Screen        | Route                                       | Purpose                        |
|---------------|---------------------------------------------|--------------------------------|
| Dashboard     | <a href="/dashboard">/dashboard</a>         | Home screen with overview      |
| New Report    | <a href="/report/new">/report/new</a>       | Step-by-step data entry        |
| Report Review | <a href="/report/review">/report/review</a> | Review before submit           |
| Alert List    | <a href="/alerts">/alerts</a>               | View all alerts                |
| Alert Detail  | <a href="/alert/{id}">/alert/{id}</a>       | Detailed alert view            |
| Sync Status   | <a href="/sync">/sync</a>                   | Pending reports, sync progress |
| Settings      | <a href="/settings">/settings</a>           | Language, about                |

## Screen Flow Diagram



## Screen 1: Login Screen (</login>)

**Purpose:** Authenticate ASHA worker

### UI Elements:

- App logo
- Phone number input (read-only for demo with pre-filled number)
- "Get OTP" button
- OTP input (simulate with "123456" for demo)
- "Login" button
- "Demo Mode" toggle (for prototype)

### Logic:

- For prototype: hardcoded OTP bypass
- Store worker ID and ward mapping
- Navigate to Dashboard

## Screen 2: Dashboard (</dashboard>)

**Purpose:** Home screen for daily workflow

## UI Elements:

- Welcome message with worker name
- **Stats Cards:**
  - Today's reports submitted
  - Pending reports (offline count)
  - Unread alerts count
- **Quick Actions Grid:**
  -  New Daily Report
  -  Maternal Risk
  -  Child Risk
  -  Environment Flag
  -  View Alerts
  -  Sync Status
- **Recent Alerts** (latest 2-3)
- **Sync Status Bar** (showing pending count)

## Design Principles:

- Large touch targets (minimum 48x48dp)
- High contrast colors
- Simple language (Marathi/Hindi/English)
- Clear visual feedback on taps

## Screen 3: New Report Flow (Multi-step)

Based on the systematic data collection process in existing apps :

### Step 1: Syndromic Surveillance

 Step 1/4: Symptom Counts

|                                                                                     |              |       |                                                                                     |
|-------------------------------------------------------------------------------------|--------------|-------|-------------------------------------------------------------------------------------|
|  | Fever Count: | [ 5 ] |  |
|  | Cough Count: | [ 3 ] |  |
|  | Diarrhea:    | [ 2 ] |  |
|  | Jaundice:    | [ 0 ] |  |

[Other Symptoms (optional)]

[Next →]

### Step 2: Maternal & Child Health

 Step 2/4: Maternal & Child

|                      |       |
|----------------------|-------|
| Pregnant Women:      | [ 2 ] |
| High Risk Pregnancy: | [ 0 ] |

ANC Visits Done: [ 1 ]

Children Under 5: [ 8 ]

Malnourished: [ 1 ]

Immunization Due: [ 3 ]

[← Back] [Next →]

### Step 3: Environmental

 Step 3/4: Environment

☒ Stagnant Water? [Yes/No]

☒ Poor Sanitation? [Yes/No]

☒ Garbage Dumping? [Yes/No]

Other Risks: [\_\_\_\_\_]

[← Back] [Next →]

### Step 4: Location & Review

 Step 4/4: Location

 Current Location Captured

Ward: Sadar Bazaar

Accuracy: 10m

[Recapture Location if needed]

[Review Full Report]

[Save Draft] [Submit]

### Key Design Decisions:

- **Progress indicator** shows step position
- **Large +/- buttons** for count adjustment
- **Default values** from previous day (optional)
- **Validation** on each step
- **Save draft** for offline

### Screen 4: Report Review

**Purpose:** Review all data before final submission

### UI Elements:

- Summary cards for each section
- Edit buttons per section
- "Submit Now" button
- "Save for Later" button

## Screen 5: Alert List (/alerts)

**Purpose:** View all received alerts

### UI Elements:

- List of alerts with:
  - Severity color (red/orange/blue)
  - Title
  - Date
  - Brief description
- Tap to view full details
- Unread indicator

## Screen 6: Sync Status (/sync)

**Purpose:** Manage offline reports

### UI Elements:

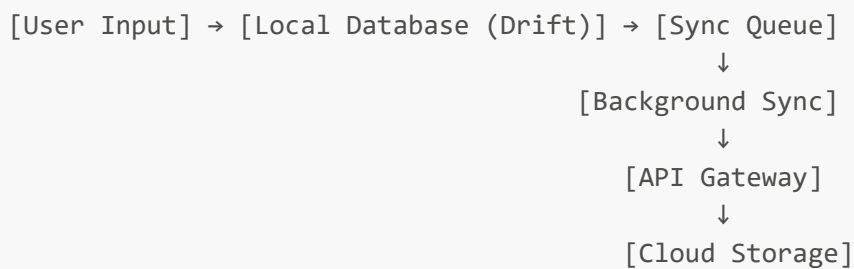
- Pending reports count
- List of pending reports with dates
- "Sync Now" button
- Last sync timestamp
- Individual report retry option

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## 6. Offline-First Architecture

Based on the critical requirement from field workers :

### Data Flow



### Sync Strategy

### 1. On Report Save:

- Save to local Drift database with `syncStatus = pending`
- Add to sync queue
- Show in pending list

### 2. On Network Available:

- Background sync worker runs
- Processes queue FIFO
- On success: mark as `synced`
- On failure: retry with exponential backoff

### 3. Conflict Resolution:

- Server timestamp wins
- If conflict detected, fetch latest from server and merge intelligently

## Local Database Schema (Drift)

```
// drift file
import 'package:drift/drift.dart';

class Reports extends Table {
  TextColumn get reportId => text().unique();
  TextColumn get ashaWorkerId => text();
  TextColumn get wardCode => text();
  DateTimeColumn get reportDate => dateTime();
  DateTimeColumn get submissionTime => dateTime();
  IntColumn get syncStatus => integer(); // 0=pending, 1=synced, 2=error

  // Syndromic data stored as JSON
  TextColumn get syndromicData => text();
  TextColumn get maternalData => text();
  TextColumn get childData => text();
  TextColumn get environmentalData => text();

  // Location
  RealColumn get latitude => real();
  RealColumn get longitude => real();

  @override
  Set<Column> get primaryKey => {reportId};
}
```

---

## 7. Push Notification Flow

**Purpose:** Receive outbreak alerts and advisories from MHO

## Implementation:

- Firebase Cloud Messaging (FCM)
- Subscribe to `ward_{wardCode}` topic on login
- Receive notifications for ward-specific alerts
- Store alerts in local database
- Show notification badge on dashboard

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## 8. Key Lessons from Poshan Tracker Reviews

### What We MUST Avoid

| Problem                  | Our Solution                                                                                  |
|--------------------------|-----------------------------------------------------------------------------------------------|
| <b>Frequent crashes</b>  | Comprehensive error handling, crash reporting (Firebase Crashlytics), testing on real devices |
| <b>GPS failures</b>      | Manual location entry fallback, cache last known location, retry mechanism                    |
| <b>Slow performance</b>  | Lazy loading, pagination, optimized queries, <40MB app size                                   |
| <b>Complex UI</b>        | Large buttons, minimal text, step-by-step wizards, icon-based navigation                      |
| <b>Old device issues</b> | Test on Android 5.0+ devices, use lightweight dependencies                                    |
| <b>Data heavy</b>        | Compress images, batch sync, use protobuf or compact JSON                                     |
| <b>Poor updates</b>      | Regression testing before release, staged rollouts                                            |

### What Users Want

- **Simplicity** – "designed for real people, not computer experts"
- **Large buttons** – "clean interface, buttons are big"
- **Offline capability** – "works offline in some areas"
- **Fast data entry** – "update a child's weight and move on"
- **Visual feedback** – "color-coded risk levels"

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## 9. Implementation Priority (13 Days)

| Priority                | Screens/Features                                                                       | Day Allocation |
|-------------------------|----------------------------------------------------------------------------------------|----------------|
| <b>P1 (Must Have)</b>   | Project setup, Drift database, Login, Dashboard, New Report form (basic), offline save | Days 1-5       |
| <b>P2 (Should Have)</b> | Multi-step wizard, sync queue, background sync, API integration                        | Days 6-8       |

| Priority                 | Screens/Features                                         | Day Allocation |
|--------------------------|----------------------------------------------------------|----------------|
| <b>P3 (Nice to Have)</b> | Alert list, push notifications, multi-language, settings | Days 9-11      |
| <b>Buffer</b>            | Testing, bug fixes, video recording                      | Days 12-13     |

### Day-by-Day Breakdown

| Day | Focus                                              |
|-----|----------------------------------------------------|
| 1   | Project setup, architecture, Drift database schema |
| 2   | Login screen, dashboard UI                         |
| 3   | New Report – Step 1 (Syndromic) with offline save  |
| 4   | Steps 2-3 (Maternal/Child/Environmental)           |
| 5   | Step 4 (Location) and Report Review                |
| 6   | Sync queue implementation, background worker       |
| 7   | API integration (POST reports)                     |
| 8   | Alert list screen, basic notifications             |
| 9   | Multi-language (Marathi translations)              |
| 10  | Polish, error handling, loading states             |
| 11  | Testing on multiple devices/emulators              |
| 12  | Bug fixes, performance optimization                |
| 13  | Final testing, video recording                     |

## 10. Integration with Arogya-SMC Backend

### API Endpoints

| Endpoint                          | Method | Purpose                           |
|-----------------------------------|--------|-----------------------------------|
| <a href="#">/api/asha/login</a>   | POST   | OTP verification (demo: simulate) |
| <a href="#">/api/asha/report</a>  | POST   | Submit completed report           |
| <a href="#">/api/asha/pending</a> | GET    | Get pending reports (for sync)    |
| <a href="#">/api/asha/alerts</a>  | GET    | Fetch alerts                      |
| <a href="#">/api/ward/{code}</a>  | GET    | Get ward details                  |

### Authentication Flow (Prototype)

For demo purposes, simplify:

1. Hardcoded worker list in `users.json`
2. "Login" accepts any 6-digit number ending with registered phone
3. JWT token returned (simulated)
4. Store token securely

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## 11. Project Structure

```
lib/  
├─ main.dart  
├─ app.dart  
├─ core/  
│   ├── constants/  
│   ├── themes/  
│   └── utils/  
├─ data/  
│   ├── models/  
│   │   ├── report.dart  
│   │   ├── alert.dart  
│   │   └── user.dart  
│   ├── database/  
│   │   ├── database.dart (Drift)  
│   │   ├── report_dao.dart  
│   │   └── sync_queue_dao.dart  
│   ├── repositories/  
│   │   ├── report_repository.dart  
│   │   ├── alert_repository.dart  
│   │   └── auth_repository.dart  
│   └── services/  
│       ├── api_service.dart  
│       ├── sync_service.dart  
│       └── location_service.dart  
├─ providers/  
│   ├── report_provider.dart  
│   ├── auth_provider.dart  
│   └── sync_provider.dart  
├─ views/  
│   ├── splash/  
│   ├── login/  
│   ├── dashboard/  
│   ├── report/  
│   │   ├── step1_syndromic.dart  
│   │   ├── step2_maternal.dart  
│   │   ├── step3_environment.dart  
│   │   ├── step4_location.dart  
│   │   └── report_review.dart  
│   ├── alerts/  
│   └── settings/  
└─ l10n/ (localization files)
```

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## 12. Success Criteria for Demo Video

The 4-5 minute video should clearly show:

1. **Login** – Simple OTP/demo login
  2. **Dashboard** – Overview of today's tasks
  3. **New Report Flow** – Step through all 4 screens entering data
  4. **Offline Indicator** – Show report saved locally
  5. **Sync** – When online, report syncs (visual feedback)
  6. **Alert Reception** – Show notification and alert list
  7. **Language Switch** – (Optional) Toggle to Marathi
- 

## Conclusion

The ASHA app design incorporates lessons learned from existing government apps like Poshan Tracker , avoiding common pitfalls while building a lightweight, offline-first, user-friendly interface. By focusing on the core data collection needs—syndromic surveillance, maternal/child risk flags, and environmental observations—and implementing robust offline sync, this app will serve as a reliable tool for Solapur's frontline health workers while integrating seamlessly with the broader Arogya-SMC ecosystem.